RYTHUMITRA: A MULTILINGUAL VOICE-FIRST AI FARMING COMPANION

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***Abstract*—Abstract—Agriculture is one of the most important factors contributing to the development of a country. However, farmers face various difficulties such as delayed crop disease diagnosis,lack of access to government schemes, and lack of information on market and weather conditions. The current digital platforms available to farmers are fragmented and do not offer a unified solution. In this context, this paper proposes a multilingual voice assistant, called "RythuMitra," which will act as a companion for farmers. The proposed system is a unified intelligent platform that includes a crop disease diagnosis module using a CNN algorithm. The proposed system will allow farmers to detect crop diseases and suggest fertilizers and pest control measures by uploading images of affected crop leaves. Additionally, a blockchain technology-based "Easy Apply" module is proposed, which will help farmers to apply schemes offered by the government. The proposed system includes several features such as crop disease detection, "Easy Apply," market price information, weather-based farming information, and a voice assistant-based farm diary.The proposed system is developed using a web-based platform with HTML, CSS, and JavaScript as programming tools. The proposed intelligent platform is experimentally evaluated, and it is shown that the proposed platform is helpful to farmers and can be used to develop smart and sustainable agriculture.**

***Index Terms*—Smart Agriculture, Crop Disease Detection, Convolutional Neural Networks (CNN), Blockchain Technology, IPFS, Government Scheme Automation, Precision Farming, Farmer Support Systems.**

I. Introduction

Agriculture is one of the most important industries that have greatly helped the world progress towards food security and economic growth, especially in developing countries where the majority of the population depends on this sector for their livelihood. Even after the advent of technology, the farmers are facing many problems, including the late detection of diseases in the crops, lack of transparency in the government schemes, lack of knowledge about the price fluctuations, and lack of awareness about the agricultural practices based on the weather conditions.

In recent times, new technologies like Artificial Intelligence (AI), Internet of Things (IoT), and Blockchain have also shown significant promise in resolving different issues in the agricultural domain. AI-based image classification approaches, especially using Convolutional Neural Network (CNN) models, have been successful in identifying diseases from leaf images using image classification. Similarly, blockchain technology can also be used for transparent record management, which can prevent fraudulent activities. Currently, many different agricultural-based applications are addressing different aspects of agricultural needs, like disease identification, weather forecasting, or even information about market prices. This is causing fragmentation in agricultural-based applications, which is making it difficult for farmers to use different applications for different needs.To overcome the aforesaid limitations, this paper proposes a multilingual voice-first AI farming companion, namely RythuMitra, which incorporates several smart agriculture services into a single platform. The proposed system will include a combination of crop disease detection through AI, as well as government scheme management through blockchain, thereby providing a complete solution for the farming community. The proposed system will allow the farmer to upload the leaf image of the crop, which will be used for real-time detection of diseases through a CNN-based model. Furthermore, the proposed system will also provide fertilizer and pest control recommendations for the crops. In addition, the proposed system will also include a blockchain-based "Easy Apply" feature, which will be used for the submission of government scheme applications through the use of smart contracts and IPFS for decentralized data storage.

Moreover, the proposed system will also include several other features, such as real-time price updates, weather-based farming advisories, and a voice-based farm diary, which will be used for recording observations in different languages. The proposed system will help in the advancement of smart and sustainable agriculture.

II. Existing Systems

The existing agricultural applications are mostly focused on delivering general information regarding crop cultivation, weather, and other market-related information. The problem is that many of these systems are based on text-based interfaces, which are often difficult for farmers who are not highly literate or technologically savvy. Therefore, the overall usability of these systems is quite poor, especially for farming communities.Several research studies have been done regarding deep learning for plant disease detection and chatbots for answering queries. Even though these systems are quite promising, they are often limited in scope. Many of these systems are often tested for a small number of crops, are usually implemented in a single language, or are often tested in a controlled environment.

In spite of achieving good accuracy for detecting diseases in crops, their efficiency is often hampered when applied in practical situations. This is because of environmental issues like improper lighting, damaged or partially exposed leaves, complex backgrounds, and environmental noise. Similarly, many agricultural chatbots are also based on rule-based systems and are not capable of understanding regional dialects, regional agricultural words, or multilingual voice interactions.

In addition, access to government schemes and subsidies is also a major problem for farmers. This is because accessing government schemes requires a complex documentation system, which is often time-consuming. This is also because many of these schemes are not accessible digitally. This is also a major problem for farmers, who are often not able to access government schemes.

The above discussion indicates that many of these systems are not capable of providing a platform that can integrate AI-based crop disease detection, multilingual voice interaction, agricultural advisory services, and secure scheme processing. This indicates a need for a system that can solve many of these agricultural problems within a single platform.

III. Proposed System

To overcome the limitations of existing agricultural platforms, this work proposes a multilingual voice-first AI farming companion, named RythuMitra, which combines intelligent crop monitoring, agricultural advisories, and transparent government scheme processing within a single digital platform. The proposed system is designed to help farmers make informed decisions in agriculture by utilizing advanced technologies like Artificial Intelligence, Blockchain, and Web services.

The core element of the system includes an artificial intelligence-based crop disease detection module, which uses Convolutional Neural Networks (CNN) for image analysis of the crops' leaves, which are uploaded by the farmers. The model performs image preprocessing, feature extraction, and classification, which help in identifying the potential diseases in crops such as rice, corn, and cotton, among others. Depending on the identified disease, the system offers recommendations on fertilizer use, pest control, irrigation, and other steps that can be taken in agriculture.

To increase transparency and facilitate easy access for the farmers in government welfare programs, the system includes a blockchain-based scheme application module. The module offers a facility called 'Easy Apply,' which can be used by the farmers to apply for government schemes related to agriculture. The details are stored using InterPlanetary File System (IPFS) technology, and the transaction occurs on a blockchain network with the help of smart contracts.

Apart from the detection of diseases and the management of schemes, the proposed system also offers real-time agricultural advisory services. This includes the provision of real-time mandi prices, weather-based farming suggestions, and crop-specific notifications, which can be extremely helpful for the farmers to efficiently schedule their farming and harvesting activities. In addition to this, the proposed system also offers a voice-based farm diary option for the farmers to effectively record their observations by providing the option to input voice-based information in their native language.

The proposed system is designed by utilizing a web-based system architecture. This includes the provision of a front-end interface for the farmers to effectively use the various services offered by the proposed system. In this regard, the proposed system can be effectively implemented by utilizing the benefits offered by AI-based technologies, blockchain technologies, and the multilingual capabilities offered by the proposed system. This can help the proposed system to effectively provide a comprehensive digital system for the farmers and the advancement of smart agriculture.

IV. Working

The proposed system, namely *RythuMitra*, utilizes Artificial Intelligence and Blockchain technologies to bring a unified digital platform to assist farmers in crop monitoring, disease detection, and government scheme access. The proposed system will follow a systematic flow of activities such as image processing, document handling, eligibility checking, and blockchain-based application handling.

The proposed system will begin with a simple dashboard-based interface through which farmers can access the platform via a web or mobile device. The proposed system will allow users to upload images, documents, and interact through voice or text inputs. The uploaded crop images will be fed into the proposed system, and the images will be preprocessed by applying various techniques such as resizing, normalization, and RGB conversion.

The images are then fed into a Convolutional Neural Network (CNN) model, which has been trained on multiple crop disease datasets. The CNN model performs a classification task to detect the occurrence of plant diseases and predict

the probability of occurrence of each disease category. Based on the prediction results, the system recommends fertilizers, pesticides, and pest control measures to the farmers to take corrective action. Besides plant disease detection, the proposed platform offers a blockchain-based platform to apply government schemes. The platform allows farmers to upload necessary documents such as Aadhar card, land documents, and bank details. The documents are stored securely using the InterPlanetary File System (IPFS). The platform verifies the eligibility criteria and enables farmers to apply through smart contracts deployed on a blockchain network. The proposed platform also offers additional features such as irrigation information, market information, and weather information through a conversational interface. The proposed platform combines crop intelligence using AI and document security using blockchain to offer a reliable platform for farmers.

A. Problem Definition

Farmers usually encounter issues such as the delay in identifying the diseases affecting crops, the unavailability of reliable information related to agriculture, and the complicated procedure for applying to government welfare schemes. Existing agricultural support systems offer fragmented services and possess text-based interfaces that are not easily accessible for farmers. Moreover, the existing system for disease detection may not work efficiently under real-world conditions due to factors like changes in lighting conditions, leaf damage, and the presence of noise. Hence, an integrated system needs to be designed to provide user-friendly services for the accurate detection of diseases.

Therefore, there is a need to develop an integrated and user-friendly system that can assist farmers in detecting crop disease at an early stage, provide real-time agricultural advice, facilitate multilingual voice conversation, and ensure transparent and secure access to various schemes offered by the government. By solving these challenges, agricultural productivity will be increased and farmers will be empowered.

B. System Architecture

The proposed system design comprises a multi-layered architecture that includes a frontend interface, backend services, AI-based disease detection module, and blockchain-based scheme processing module. The farmers interact with the proposed system using a web interface that enables farmers to upload crop images, access market information, and apply for government schemes. The backend server processes the user requests and interacts with the AI module for disease prediction. The proposed system also comprises a blockchain module for storing scheme information using smart contracts, while documents are stored using decentralized storage with the help of IPFS.

C. Dataset Description

The crop disease detection model is trained using a dataset containing images of crop leaves obtained from publicly available agricultural data sets. The dataset consists of various

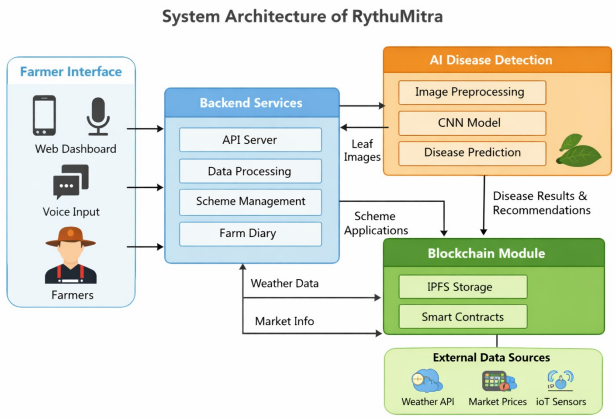

Fig. 1. System Architecture of the Proposed RythuMitra Platform

crop disease types for different crops such as rice, corn, and cotton. The images are labeled according to the respective disease type or normal condition. Before training the model, the images are subjected to various preprocessing steps such as resizing, normalization, and removal of noise to maintain consistency in the input data to the model. Data augmentation methods are also employed to increase the diversity of the dataset and enhance the robustness of the model under real-world crop field conditions.

D. AI-Based Disease Detection Model

The system uses a Convolutional Neural Network (CNN) model for the classification of plant diseases. In this model, the CNN architecture uses leaf images, and the features are extracted using convolution and pooling layers. In the next stage, the features are classified using fully connected layers with a softmax activation function. In the final stage, the leaf image, which is uploaded by the user, is processed by the model, and the most probable disease class with a confidence level is predicted. According to the predicted disease, the system provides recommendations on fertilizer use, pest control, irrigation, and prevention methods.

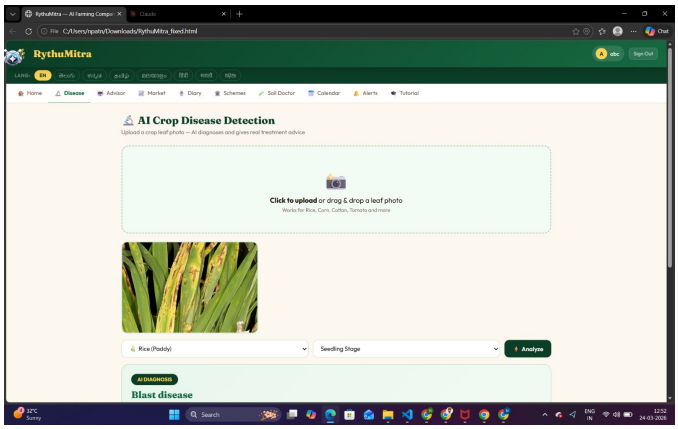

Fig. 2. Disease Detection Using CNN Model

E. Model Training and Evaluation

The crop disease detection module is developed based on a Convolutional Neural Network (CNN) model, which is implemented through the TensorFlow and Keras platforms. The dataset for training the model is a set of labeled images of crop leaves, which are either healthy or have diseases. The model is designed for the classification of diseases that affect crops such as rice, corn, and cotton.

During the training process, the dataset is divided into training and validation sets, which are used for testing the generality of the model. Preprocessing techniques for the images, such as resizing, normalizing, and augmenting, are used for enhancing the robustness of the model and for preventing overfitting. The techniques used for augmenting the images include rotation, flipping, and scaling.

In the CNN model, the hierarchical features are extracted from the leaf images using the convolutional layers, followed by the pooling layers, which reduce the spatial dimensions of the images while retaining the important information. Finally, the classification layer uses the softmax activation function to classify the images into the probability of the disease classes.The performance of the proposed model is evaluated using the performance metrics, which determine the effectiveness of the model in identifying the crop diseases.

F. AI–Blockchain Integration

The proposed system utilizes Artificial Intelligence and Blockchain technologies for an integrated solution for the agriculture sector. The Artificial Intelligence module is responsible for crop image analysis and detection of diseases, while the Blockchain module is responsible for handling government scheme applications in a transparent and tamper-free manner.

When a farmer uploads an image of a crop, the server receives the image and sends it for prediction to the pre-trained CNN model. The results are sent back to the application, along with recommendations for the farmer. Simultaneously, the system also provides a facility for the farmer to apply for government schemes.

To ensure transparency and prevent any tampering of government scheme application data, the documents are stored on the InterPlanetary File System (IPFS), which provides a decentralized and tamper-free data storage solution. The IPFS hash is stored on the Blockchain through smart contracts, which are deployed on a test network.

G. Blockchain-Based Scheme Processing

To ensure transparency and trust in government schemes' applications, a blockchain-based module for processing the schemes' applications has been included in the system. When a farmer applies for a scheme, the documents are uploaded to a decentralized storage system using the IPFS. The hash of the uploaded documents in the decentralized storage system is recorded on a blockchain network using a smart contract transaction. This ensures that the documents uploaded for a scheme's application are immutable, verifiable, and tamper-proof.

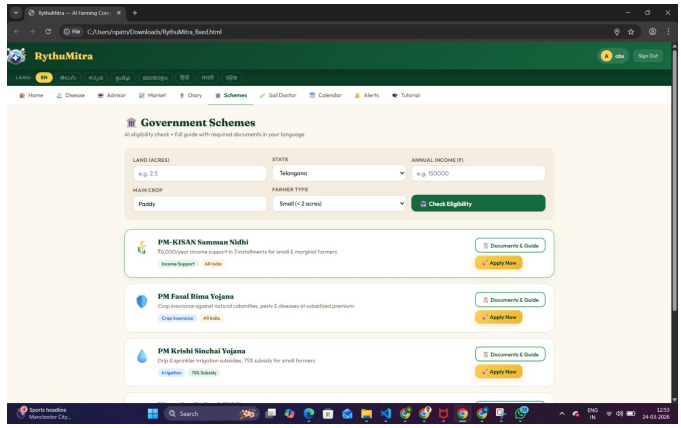

Fig. 3. Blockchain-Based Government Scheme Processing

H. User Interface Design

The user interface of the proposed system is simple and user-friendly. The user interface of the proposed system is such that it can be used by farmers of varying knowledge. The dashboard of the proposed system is such that it can be used to access all the services offered by the proposed system. These services include crop disease detection, market price information, weather information, government schemes, and farm diary.

The proposed system includes a multi-language interface. The multi-language interface of the proposed system is such that it can be used by farmers of different areas. The proposed system includes voice-based input functions. The proposed system is such that it can be used by farmers even if they are not well-versed in using digital technologies. The proposed system includes icons and visual information. The proposed system is such that it can be used by farmers to efficiently use the features of the proposed system.

I. Workflow of the System

The overall workflow of the system starts from the point where the farmer enters the system through the dashboard interface. The farmer can perform various tasks such as uploading images for disease detection, making observations through voice input, checking prices and weather information, and applying for government schemes. The backend receives the request from the farmer and communicates with the corresponding modules, which may include the AI model or the blockchain network. The system returns the response to the user in the form of predictions, recommendations, or confirmations for the schemes. This is the overall workflow of the system.

J. Web-Based Deployment

To ensure that the system is accessible and user-friendly for farmers, the proposed system is implemented as a web-based application that can be accessed through a web browser on desktop or mobile devices. The user interface is designed to

be user-friendly and easy to interact with for farmers who may not have any technical expertise.

The frontend of the proposed system is implemented using HTML, CSS, and JavaScript programming languages to create a user interface that enables farmers to interact with multiple agricultural services through a single interface. Using the interface, farmers can upload crop images for disease detection, access real-time market price information, check weather conditions, record observations using voice input, and apply for government schemes.

The backend services are developed using a server based on Node.js. This server handles API requests from the frontend and facilitates communication with various system modules. The modules include an AI-based disease detection model, a blockchain-based scheme processing system, and API services for obtaining weather and market details.

When a user wants to upload an image of their crop, this image is sent to the backend server. This server uses a trained convolutional neural network model to process the image and detect potential diseases. The results of this prediction and agricultural actions like fertilizers and pest control measures are then sent back to the frontend and displayed to the user in real time.

The web-based implementation offers various advantages like scalability, cross-platform compatibility, and ease of maintenance. This system also allows farmers to access their agricultural services from anywhere and anytime with internet connectivity. This makes the system very practical and useful for providing agricultural and digital services to farmers.

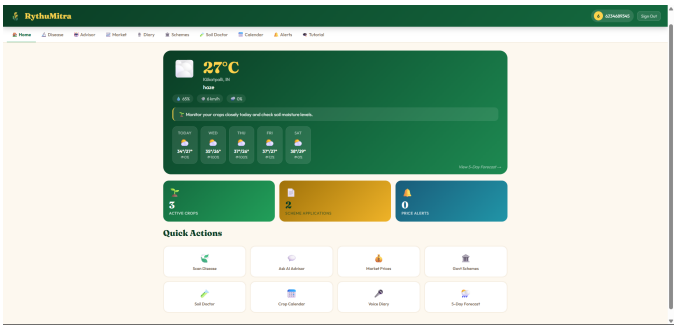

Fig. 4. Web-Based Dashboard Interface of RythuMitra

V. Results

The proposed RythuMitra system was evaluated to assess the efficacy of the crop disease detection module using the AI algorithm and the overall performance of the combined platform. The proposed system was evaluated by using sample crop leaf images representing different crop diseases of crops such as rice, corn, and cotton. The trained CNN model was able to perform well in identifying crop diseases and differentiating between healthy and diseased crop leaves.

While evaluating the proposed system, it was able to show high accuracy while classifying crop diseases using the validation dataset. The proposed model was able to show its capability of identifying crop diseases such as diseases of crops such as rice, corn, and cotton. The pre-processing and augmentation of the proposed model helped improve its robustness by allowing it to perform well even while handling different lighting conditions, leaf orientation, and background noise. The proposed model predicts crop diseases and displays the result as a pop-up window showing the identified crop disease along with its confidence score.

Another feature that was successfully implemented was the generation of recommendations for agriculture based on the determined category of diseases. This entails recommendations for fertilizer use, pest control, irrigation, and prevention. This feature will help farmers make appropriate decisions that will improve crop yields.

Apart from the AI-based detection module, the blockchain-based scheme application feature was also implemented for testing, focusing on transparency and reliability. The inclusion of IPFS for document storage was also successful, providing a decentralized and secure method for storing scheme application documents. This is combined with a blockchain layer for transaction verification through smart contracts. The transaction hash enables scheme applications to be tracked transparently.

The web-based dashboard was also tested for usability and accessibility. Farmers can upload crop images, see the results for crop disease detection, check market prices, access weather updates, and even input voice for farm observations. The integrated platform was successfully used to demonstrate the potential for integrating different services for farmers within a single user-friendly interface.

Overall, the experimental results showed that the proposed system was an effective solution for smart agriculture. The combination of crop intelligence through AI, transparency through blockchain, and multilingual access has improved the decision-making capabilities of the system, as well as the availability of digital services for farmers.

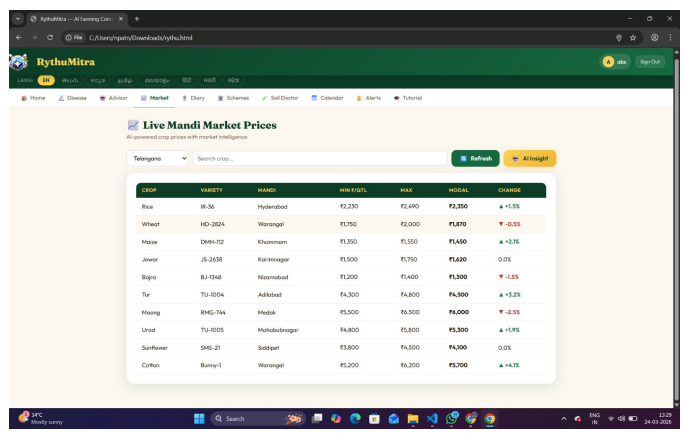

Fig. 5. Live Mandi Market Prices

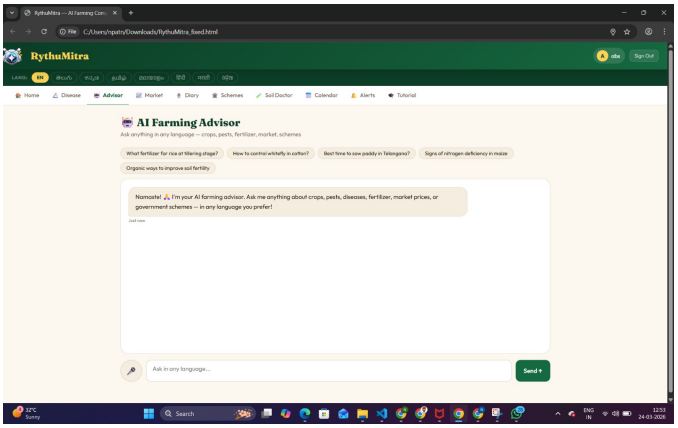

Fig. 6. Voice Driven Farm Diary

VI. Conclusion

In this paper, a multilingual voice-first AI farming companion named RythuMitra was proposed, which can help farmers through an integrated digital agriculture platform. In the proposed system, Artificial Intelligence and Blockchain technologies are used to solve a number of important problems faced by the farming community, which include early detection of crops' diseases, lack of easy access to agriculture-related advisory services, and complex procedures for applying for government schemes. In the proposed system, the crop disease detection module uses Convolutional Neural Networks (CNN) for the analysis of leaf images and accurately detects diseases in crops, which can help the farming community in early prevention of diseases in crops. Moreover, the proposed system uses Blockchain technology for the application of government schemes, which can help in transparent, secure, and tamper-proof management of records through IPFS and smart contracts.

Other features are also included in the proposed system, which can help the farming community, such as real-time weather information, live market prices, and a voice-based farm diary, where a farmer can record his observations in his native language itself. The proposed system can be accessed through a web-based interface, which can be used practically in the farming environment.

Overall, the proposed system illustrates the potential benefits of incorporating AI-based crop intelligence and blockchain transparency in the development of digital support systems for agriculture.

Future Improvements In the future, the proposed system can be improved in a number of ways, for example, by incorporating a wide range of crops and disease categories through the use of large amounts of data, as well as incorporating IoT-based sensors for monitoring soil moisture, temperature, and other environmental conditions, which can help in improving the precision agriculture support capabilities of the proposed system, as well as improving the multilingual voice support and making the system a mobile application for easy use by farmers in rural areas.

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